External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

TEACHER GUIDE · 1 OF 9

Prepare the purpose, prerequisites and materials

> Prepare External side effects and recovery: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of external side effects and recovery

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of timeouts retries and cancellation

Reference example, not the only acceptable wording: Teacher explanation: Timeouts bound waiting; retries consume a finite budget. Cancellation should stop future work while accounting for operations already committed.

Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is 3*4+1+2=15 seconds.

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of planning and action selection

Reference example, not the only acceptable wording: Teacher explanation: Planning proposes steps; execution must still validate permissions and prerequisites for each step. Replanning cannot invent authority.

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of failure budgets and service objectives

Reference example, not the only acceptable wording: Teacher explanation: Service objectives define measured reliability over a stated interval.

A failure budget is meaningful only after success and eligible requests are defined. For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are 2%*200=4.

Count every eligible failure once; report excluded requests separately.

Two 45-minute sessions. External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of external side effects and recovery

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain external side effects and recovery using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model External side effects and recovery using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

Explain External side effects and recovery · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

Explain External side effects and recovery · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Acknowledgement	A response confirming operation status
Reconciliation	Checking authoritative state after uncertainty
Compensation	A separate action intended to offset a prior effect

External side effects and recovery — evidence trace

INPUT AND SCOPE

A booking is committed in storage, but response delivery times out. Caller asks to repeat.

CHECK OR CALCULATION

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action.

CONCLUSION AND LIMIT

Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Explain External side effects and recovery · 3

Guided: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A booking is committed in storage, but response delivery times out. Caller asks to repeat.

Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: External side effects and recovery

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

The facts we will use

> **Supplied facts**
>
> Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

AI ACADEMY / AI-6/17

External side effects and recovery

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record uncertainty

Sending occurred, but printing completion is not yet known.



2. Query J6

Ask for the existing job's recorded state.



3. Recover from status

If complete, return that result without issuing another print job.

CHANGE ONE THING

Status reliably reports J6 absent.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record uncertainty	Sending occurred, but printing completion is not yet known.
2. Query J6	Ask for the existing job’s recorded state.
3. Recover from status	If complete, return that result without issuing another print job.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

What this example does not establish

Keep the stated limit

Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Change one thing in the pictured case

Change one thing

Change: Status reliably reports J6 absent.

Predicted result

Prediction: A policy-approved resend using the same deduplication key may be considered.

Why the result follows

Reason: Confirmed absence supplies different **evidence** from silence.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate External side effects and recovery, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion practice

Completion task: A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Reasoned teacher answer

Sending occurred, but printing completion is not yet known. Ask for the existing job's recorded state. If complete, return that result without issuing another print job.

Changed case: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Changed-case reasoning: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the External side effects and recovery investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the External side effects and recovery questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

W2 Trace the worked case

Workbook question · W2

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

Teacher reasoning · W4

Teacher explanation: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

The new case should test this particular **mechanism** rather than merely restate the original result.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only. Require intermediate reasoning and the stated scope.

Fresh retry: A send ledger says committed while client reports timeout. Expected reasoning: Treat the committed record as effect **evidence**; return its result and avoid duplication.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Independent question · withhold the key

Selected case: A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: No. Reconcile the operation ID with the authoritative ledger; retry only under the declared deduplication/recovery rule. Use fictional tokens only.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect the project, support and next lesson

> Connect External side effects and recovery to Grounded Learning Service, adapt support and plan a later changed-case check.

Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Untrusted **input** boundaries. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-6/18, ai-6/20, ai-6/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Prepare, teach and assess

> Use this preparation summary alongside the detailed External side effects and recovery lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of external side effects and recovery

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Check every learner before discussion: Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Status reliably reports J6 absent. Require a **prediction** and a reason before comparing. Expected result: A policy-approved resend using the same deduplication key may be considered.

Reason: Confirmed absence supplies different **evidence** from silence.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

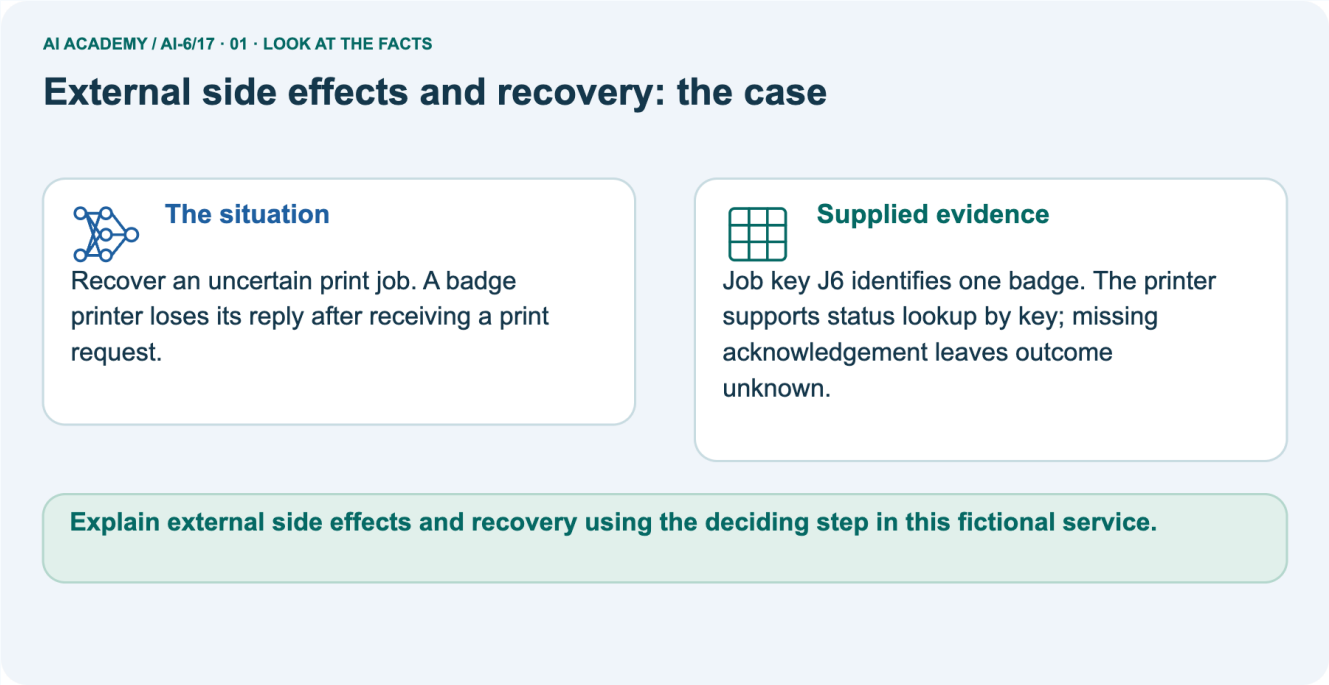
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one External side effects and recovery graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. External side effects and recovery: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Recover an uncertain print job. A badge printer loses its reply after receiving a print request. Supplied facts: Job key J6 identifies one badge.

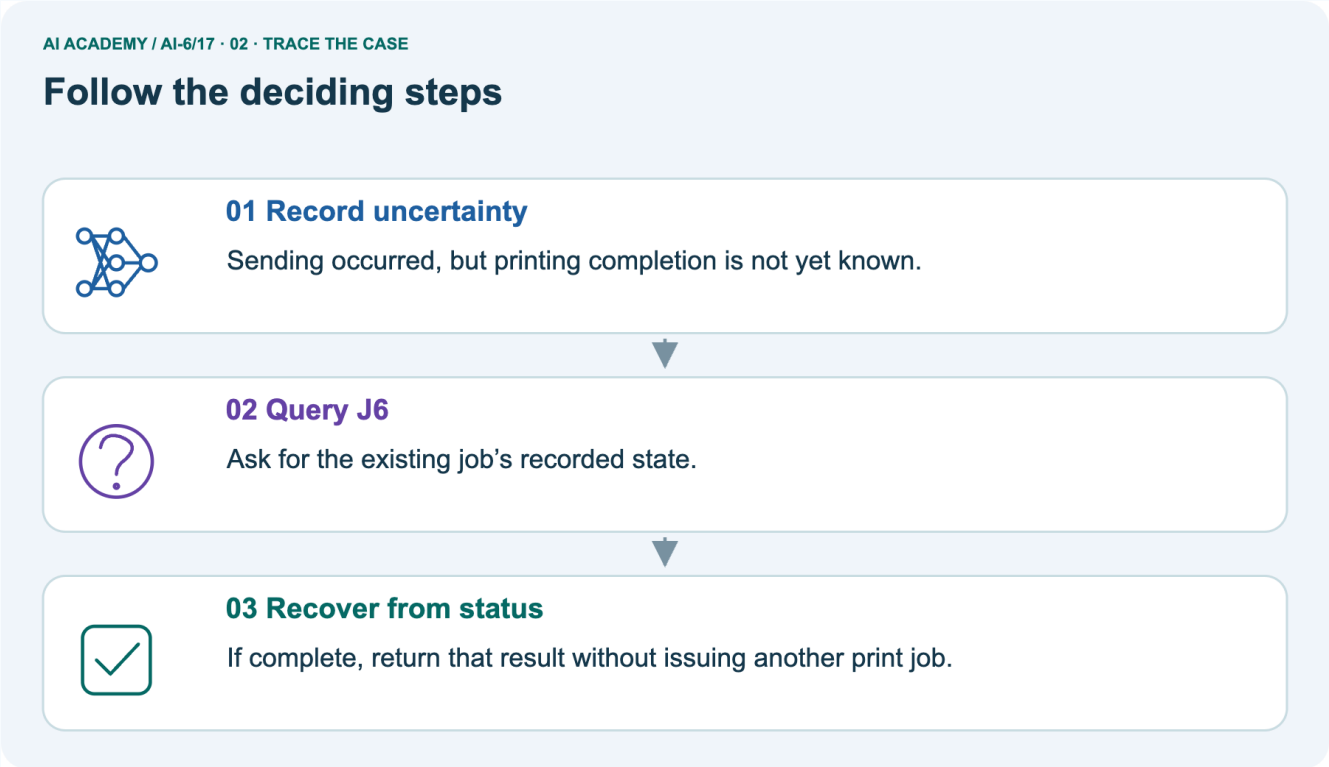
The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

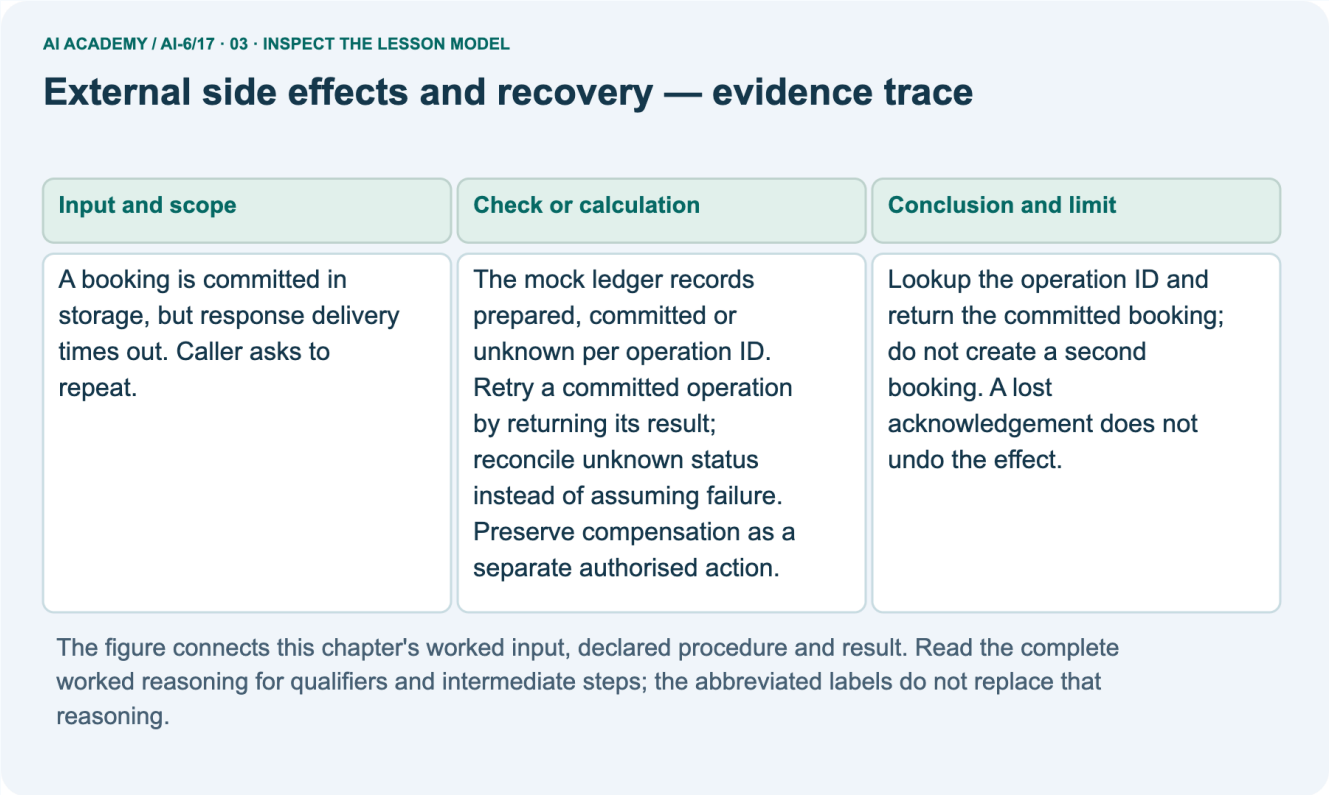
Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. External side effects and recovery — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

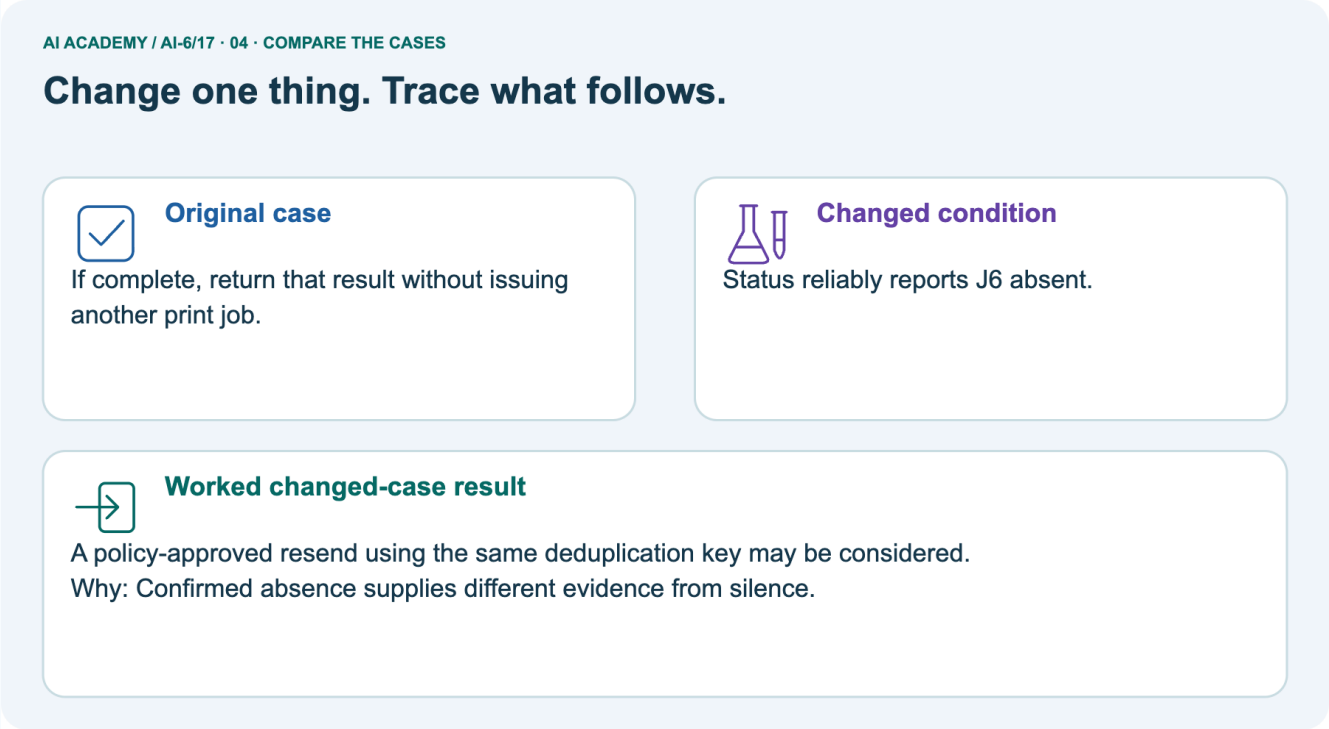
Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

> Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

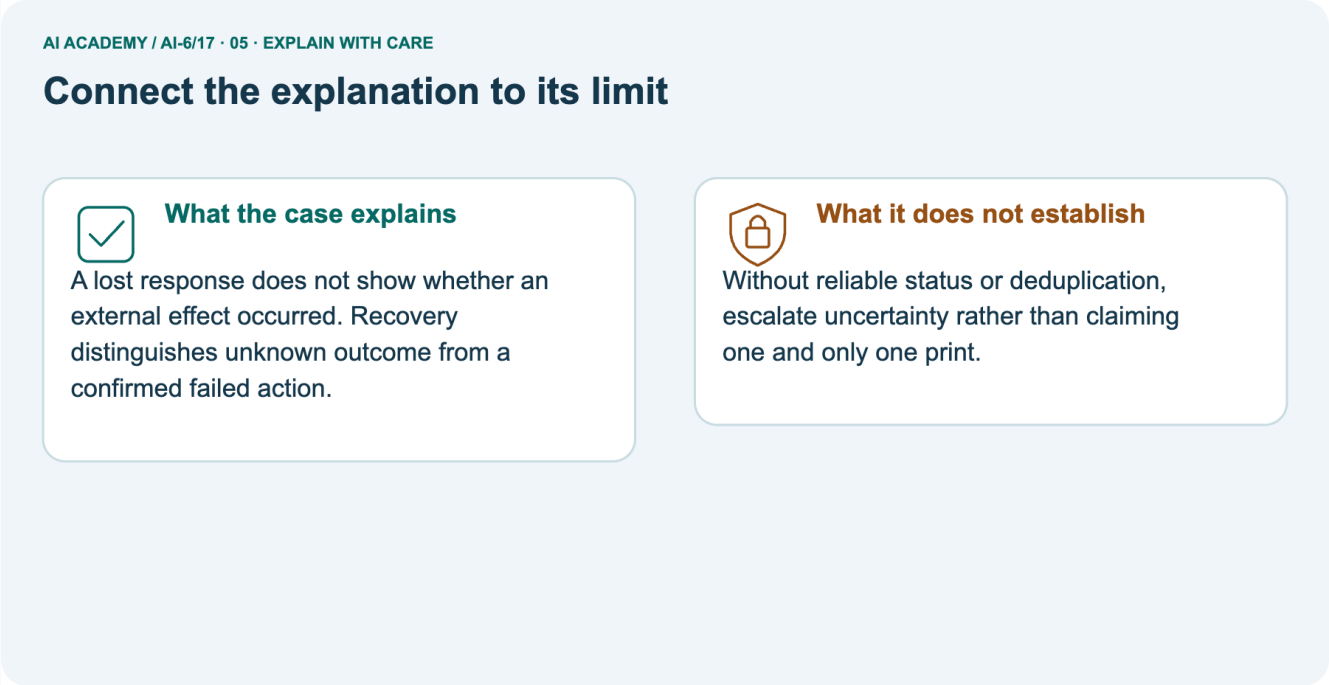
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Confirmed absence supplies different **evidence** from silence.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

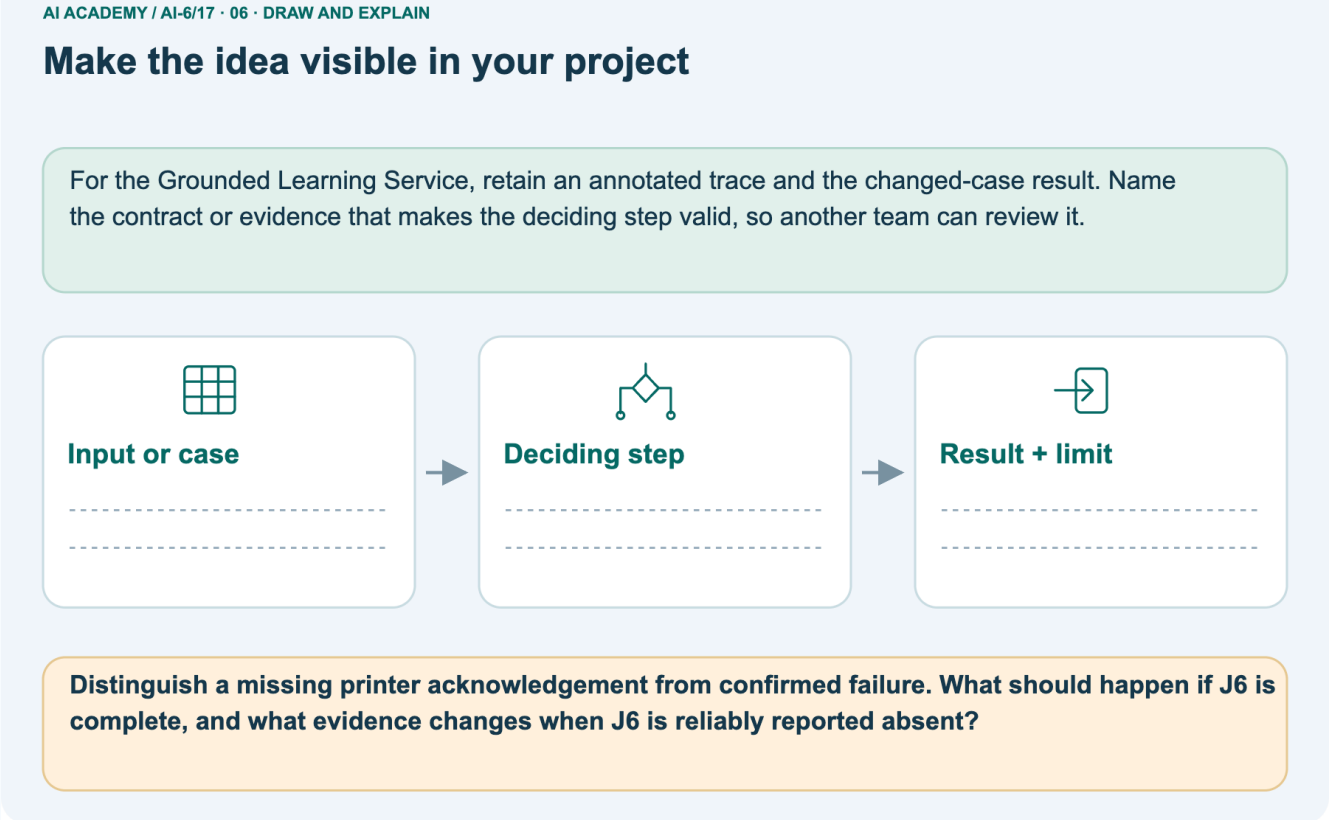
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Sending occurred, but printing completion is not yet known. Ask for the existing job's recorded state. If complete, return that result without issuing another print job.

Changed case: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.